

# General Coding Principles

## 1.1 Readability

- Code should be easy to read and understand.
- Prefer clear and simple implementations over unnecessarily complex solutions.
- Avoid writing multiple unrelated operations in a single statement.
- Use meaningful names for variables, functions, classes, and modules.
- Code should be understandable to another team member without requiring extensive explanation.

## 1.2 Modularity

- Each module should have a clearly defined responsibility.
- Functions and classes should perform one primary task.
- Avoid creating large functions containing unrelated operations.
- Common functionality should be placed in reusable modules rather than duplicated.
- Modules should interact through clearly defined interfaces.

## 1.3 Maintainability

- Code should be written with future modifications in mind.
- Avoid hard-coded values when configuration or constants are more appropriate.
- Changes to one feature should cause minimal changes to unrelated modules.
- Before adding a new feature, existing functionality should be checked to avoid duplication.

## 1.4 Reliability

- Operations that may fail should include appropriate error handling.
- External dependencies such as APIs, Windows services, files, and processes should not be assumed to always be available.
- The system should provide meaningful error messages rather than silently failing.
- Critical operations should be verified after execution whenever practical.